

Gregory Brooks

Principal Software Engineer at SuperSharp Space Systems Ltd.

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Relevant Experience

- 2026 to present **Principal Software Engineer**, *SuperSharp Space Systems Ltd.*, Cambridgeshire.
Principal software engineer developing flight software for the Hibiscus telescope.
- Established team standards for flight software development and testing (documented coding standards, defined branching & review strategy, set up CI pipelines and other automations).
 - Kept the team aligned with the state-of-the-art for space flight software development, by attending conferences like the annual Flight Software Workshop and by highlighting relevant tools, technologies and frameworks (e.g., F Prime).
- 2025 to 2026 **Senior Firmware Engineer**, *SuperSharp Space Systems Ltd.*, Cambridgeshire.
Senior firmware engineer, developing flight firmware for real-time mechatronic control of satellite deployment mechanisms.
- 2022 to 2025 **Medical Software Consultant**, *Team Consulting Limited*, Cambridgeshire.
- Software lead for the iF Design Award winning NANOME system (<https://ifdesign.com/en/winner-ranking/project/nanome/642972>).
 - Developed software for a range of applications, including:
 - Automated hardware-in-the-loop testing.
 - 3D photogrammetry.
 - User interfaces in Flutter.
 - Multi-threaded event driven firmware development in C (FreeRTOS).
 - IEC 62304 compliant software development for medical devices; I've worked as a project software lead at every level of the V-model from early stage system/software architecture to detailed design and implementation, to testing and release.
 - Performed early-stage project feasibility assessments and development proposals.
 - Applied continuous integration and delivery techniques and technologies to facilitate rapid iterative development cycles and automated testing (containerised build environments running in Bitbucket Pipelines).
 - Designed message-passing APIs using protobuf for asynchronous stream based communication between applications.
 - Developed software for bare-metal, RTOS (FreeRTOS) and Linux (Ubuntu) environments.
 - Exploratory work with Rust and Go.
 - Sat on the 2024 Winter Party planning committee.
- 2021 to 2022 **Senior Embedded Software Engineer**, *Signaloid Limited*, Cambridge.
Worked with a wide range of technologies, including:
- Software development in C and C++ for Signaloid's uncertainty-tracking processor (see github.com/signaloid for examples).
- 2019 to 2021 **Consultant (Embedded Systems)**, *TTP plc*, Melbourn.
Worked on a range of multidisciplinary projects within the Life Science/Cell and Gene team, such as:
- CoVent ventilator, a collaboration with Dyson in response to the 2020 'Ventilator Challenge' during the COVID-19 pandemic.
 - Puckdx sample-to-answer human IVD platform (for DiaSorin).
- Responsibilities included:
- Hardware (schematic and PCB) design, assembly of prototypes.
 - Firmware development for STM32 family, from bare-metal C99 to multithreaded C++11 with mBed RTOS.
 - Software/GUI development in Python (Kivy touchscreen interface for an embedded SBC).
 - Testing and debugging of hardware, firmware and software (oscilloscopes, logic analysers, SWD debuggers).
 - Implementation, testing and debugging of communications interfaces (RS485, I2C, SPI, Modbus, Ethernet<->TCP<->HTTP).
 - Communication and collaboration with other team members, especially those without an electronics/software background (e.g., scientists, project managers).
 - Project timeline estimation, prioritisation of tasks to meet aggressive deadlines.

Education

- 2015–2019 **MEng & BA Electrical and Information Sciences**, *Christ's College, University of Cambridge*, 2.1.
2008–2015 **A-level Mathematics, Further Mathematics, Physics & Chemistry**, *Sutton Grammar School, Sutton*, 4 A* grades.

Activities & Awards

- Won Google Creative Technology Prize at the national Big Bang Science Fair 2015.
- At university, I was a core member of Cambridge University Spaceflight society where I have worked on the design, construction, programming and testing/flight of projects such as an inertial measurement unit, GPS/telemetry boards for rockets, lightweight balloon payloads and a time-of-flight trilateration system for tracking a rocket's position during flight.
 - Developed Python applications and GUIs using PyQt for various society projects e.g., the trilateration project mentioned above.
 - Part of the team that launched the society's Martlet 3 rocket at Black Rock desert, Nevada, in 2017.

Master's Degree Project

- Title *Compiling Physical Invariant Descriptions to Hardware Descriptions for a Sensor Interface for Security and Privacy in IoT Applications*
- Supervisor Dr Phillip Stanley-Marbell
- Description This project involves writing a compiler, in C, that takes a description of physical laws/constraints relating electronic sensor data (e.g., pressure $\times$ volume $\propto$ temperature) and outputs Verilog RTL for use with a low power iCE40 FPGA. This FPGA sits between the sensors and external circuitry, such as a microprocessor, implementing a local differential privacy system which accounts for the physical relationships and hence mutual information between related measurements.

Degree Modules (Condensed Summary)

- Mathematics
- Signal Processing
- Information Theory
- Embedded Systems
- Analogue and Digital Electronics
- Software Engineering